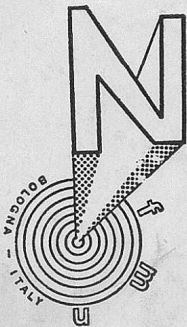


# INSTRUCTION MANUAL FOR

# THE INVANDERS



**F.lli ZACCARIA**

S.n.c.

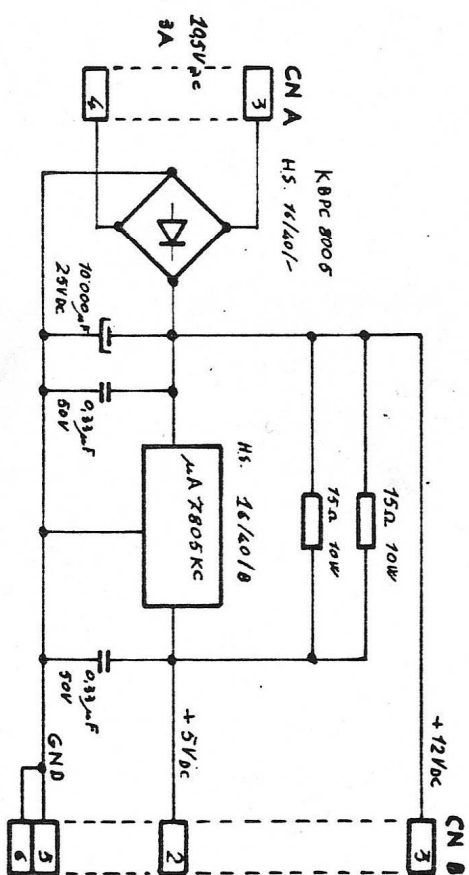
di Zaccaria Marino - Franco - Natale

**COSTRUZIONI GIOCHI D'ATTRAZIONE**

Via Ampère, 15 - 40012 CALDERARA DI RENO (Bo) Italy  
Telefono (051) 722381 / 82 con linea automatica  
Telex 810524 INTERCON

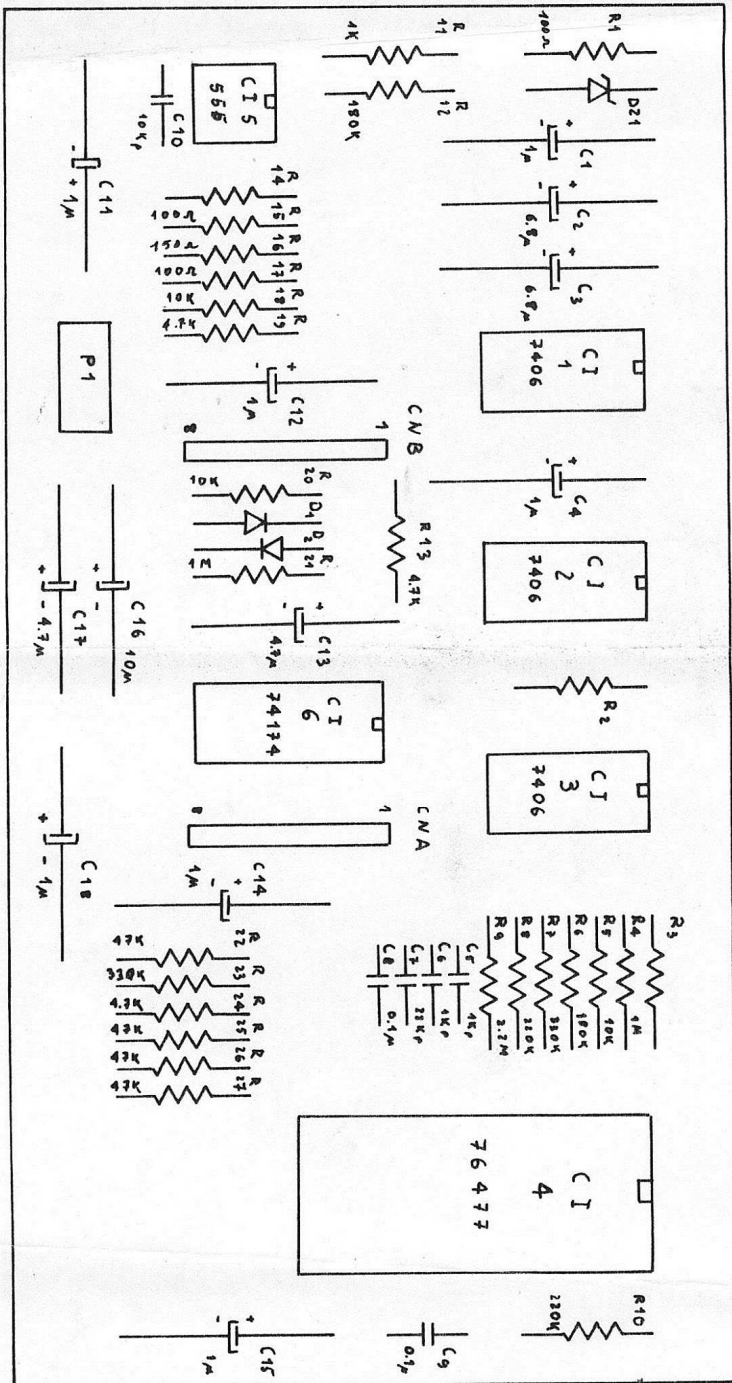
# THE INVADERS

## POWER BOARD



N.B. - Alimentatore per C.P.U. con memorie 93451  
 - Per memorie 93453 le 2 Res. da 15 $\Omega$  10W vanno sostituite con una Res. 3,9 $\Omega$  15W

# SOUND BOARD "THE INVADERS"



| Qualità | Dimensioni e scostamenti per le quote libere |       |       |       |       |       |       |       |       |       | (priva di particolari tolleranze) |       |       |       |
|---------|----------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-----------------------------------|-------|-------|-------|
|         | da 1                                         | a 3   | a 6   | a 10  | a 18  | a 30  | a 50  | a 80  | a 120 | a 180 | a 250                             | a 315 | a 400 | a 500 |
| IT 10   | 0.040                                        | 0.048 | 0.058 | 0.070 | 0.084 | 0.100 | 0.120 | 0.140 | 0.160 | 0.180 | 0.200                             | 0.220 | 0.250 | 0.280 |
| IT 11   | 0.080                                        | 0.075 | 0.090 | 0.110 | 0.130 | 0.160 | 0.190 | 0.220 | 0.250 | 0.280 | 0.320                             | 0.360 | 0.400 | 0.450 |
| IT 12   | 0.090                                        | 0.120 | 0.150 | 0.180 | 0.210 | 0.250 | 0.300 | 0.350 | 0.400 | 0.450 | 0.500                             | 0.560 | 0.630 | 0.720 |
| IT 13   | 0.140                                        | 0.180 | 0.220 | 0.270 | 0.330 | 0.390 | 0.460 | 0.540 | 0.630 | 0.720 | 0.810                             | 0.900 | 1.000 | 1.100 |

**NACCARIA**

COD.

TRATTAMENTO

PROT. SUP.

attrezzo

scarto

peso netto

DIS. DATA

SCALA

MATERIALE

quant.

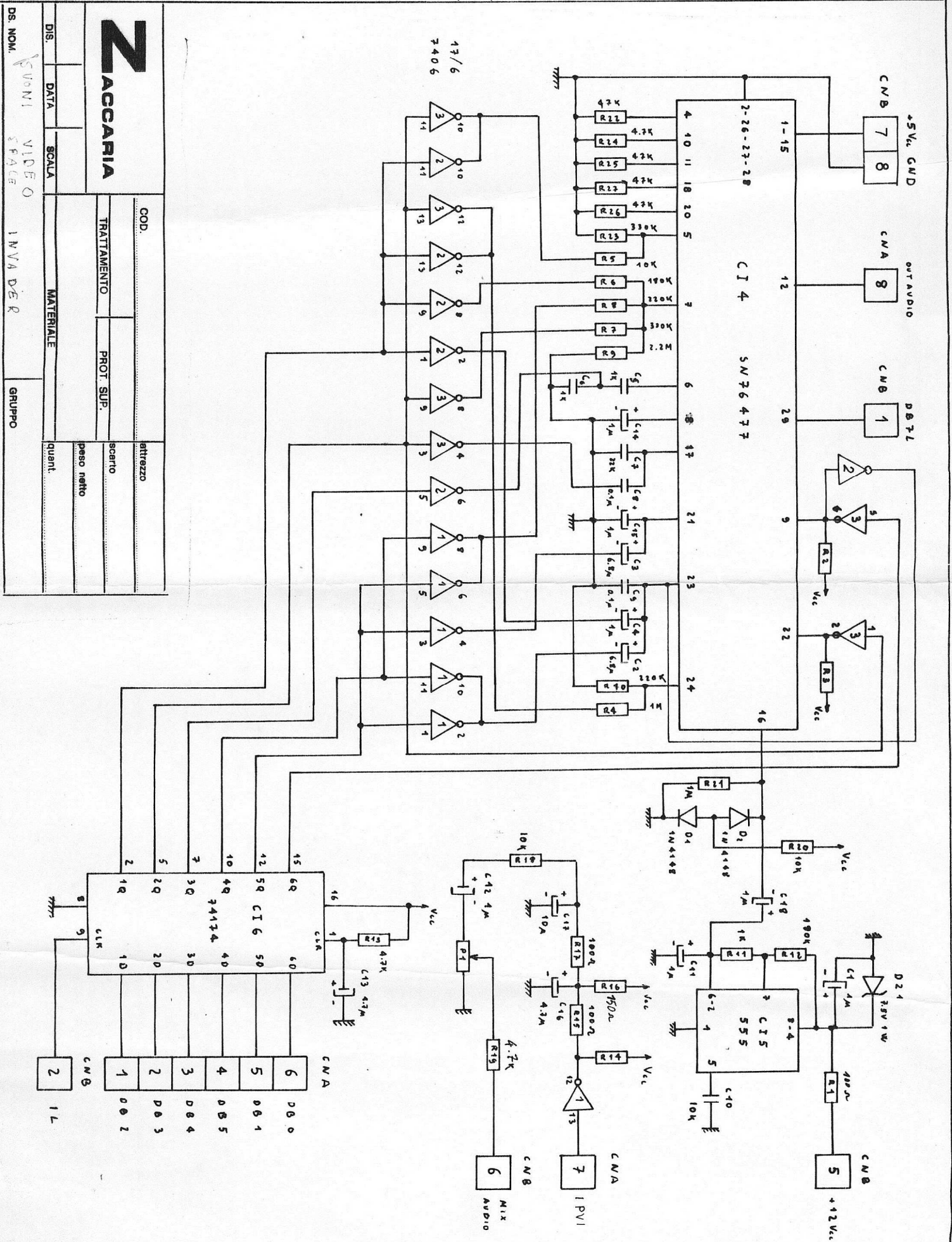
DS. NOM.

GRUPPO

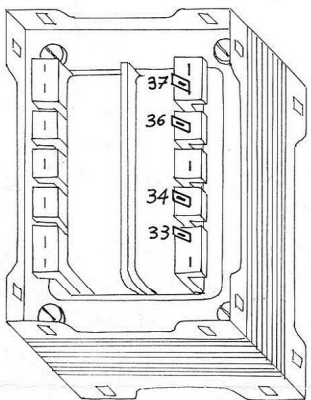
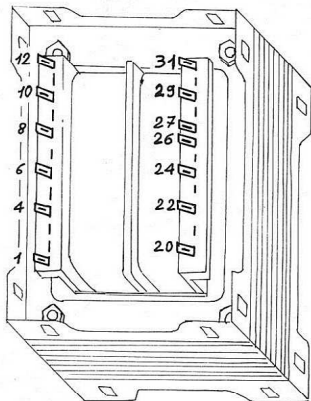
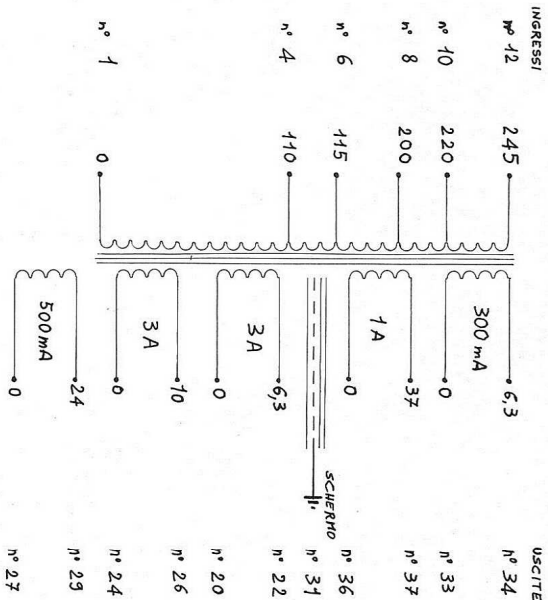
MODIFICA

FIRMA

| DATA | MODIFICA | FIRMA | Qualità | Dimensioni e scostamenti per le quote libere |       |       |       |       |       |       |       |       |       | (priva di particolari tolleranze) |       |       |       |
|------|----------|-------|---------|----------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-----------------------------------|-------|-------|-------|
|      |          |       |         | ALBERI h = da 0 a —                          |       |       |       |       |       |       |       |       |       | FORI H = da 0 a +                 |       |       |       |
|      |          |       |         | da 1                                         | a 3   | > 3   | a 6   | a 10  | a 18  | a 30  | a 50  | a 80  | a 120 | a 180                             | a 250 | a 315 | a 500 |
|      |          |       | IT 10   | 0.040                                        | 0.048 | 0.058 | 0.070 | 0.084 | 0.100 | 0.120 | 0.150 | 0.180 | 0.220 | 0.250                             | 0.290 | 0.320 | 0.360 |
|      |          |       | IT 11   | 0.060                                        | 0.075 | 0.090 | 0.110 | 0.130 | 0.160 | 0.190 | 0.230 | 0.270 | 0.300 | 0.350                             | 0.400 | 0.450 | 0.500 |
|      |          |       | IT 12   | 0.090                                        | 0.120 | 0.150 | 0.180 | 0.210 | 0.250 | 0.300 | 0.360 | 0.430 | 0.500 | 0.580                             | 0.670 | 0.760 | 0.850 |
|      |          |       | IT 13   | 0.140                                        | 0.180 | 0.220 | 0.270 | 0.330 | 0.390 | 0.460 | 0.540 | 0.630 | 0.720 | 0.810                             | 0.900 | 1.000 | 1.120 |



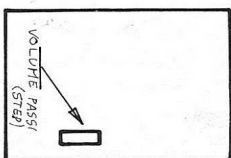
TRASFORMATORE MOD. 2 C 1003



REGOLAZIONE SEGNALE VIDEO →

OFF SWITCH  
ON 1 2 3 4 5 6 7 8

VOLUME: ESPLOSIONE BASE LASER - BERSAGLIO  
COLPITO - SPARO - UFO COLPITO - VINCITA  
BASE LASER  
VOLUME UFO →



AGGIUSTAMENTO MONETE SW1 SW2 SW3

2 COIN - 1 PLAY - OFF - OFF - OFF - OFF  
1 " - 1 " - ON - ON - OFF - OFF  
1 " - 2 " - OFF - OFF - ON - ON  
1 " - 3 " - ON - ON - OFF - OFF  
1 " - 4 " - OFF - ON - ON - OFF  
1 " - 5 " - ON - ON - ON - OFF  
1 " - 6 " - OFF - ON - ON - ON  
1 " - 7 " - ON - ON - ON - ON

N° BASI LASER SW6

3 - OFF  
4 - ON

VELOCITA' PROIETTILE INVAS. SW5

NORTH - OFF  
VELOCITA' - ON  
THROWING - ON

PUNTI SW4 BASE LASER

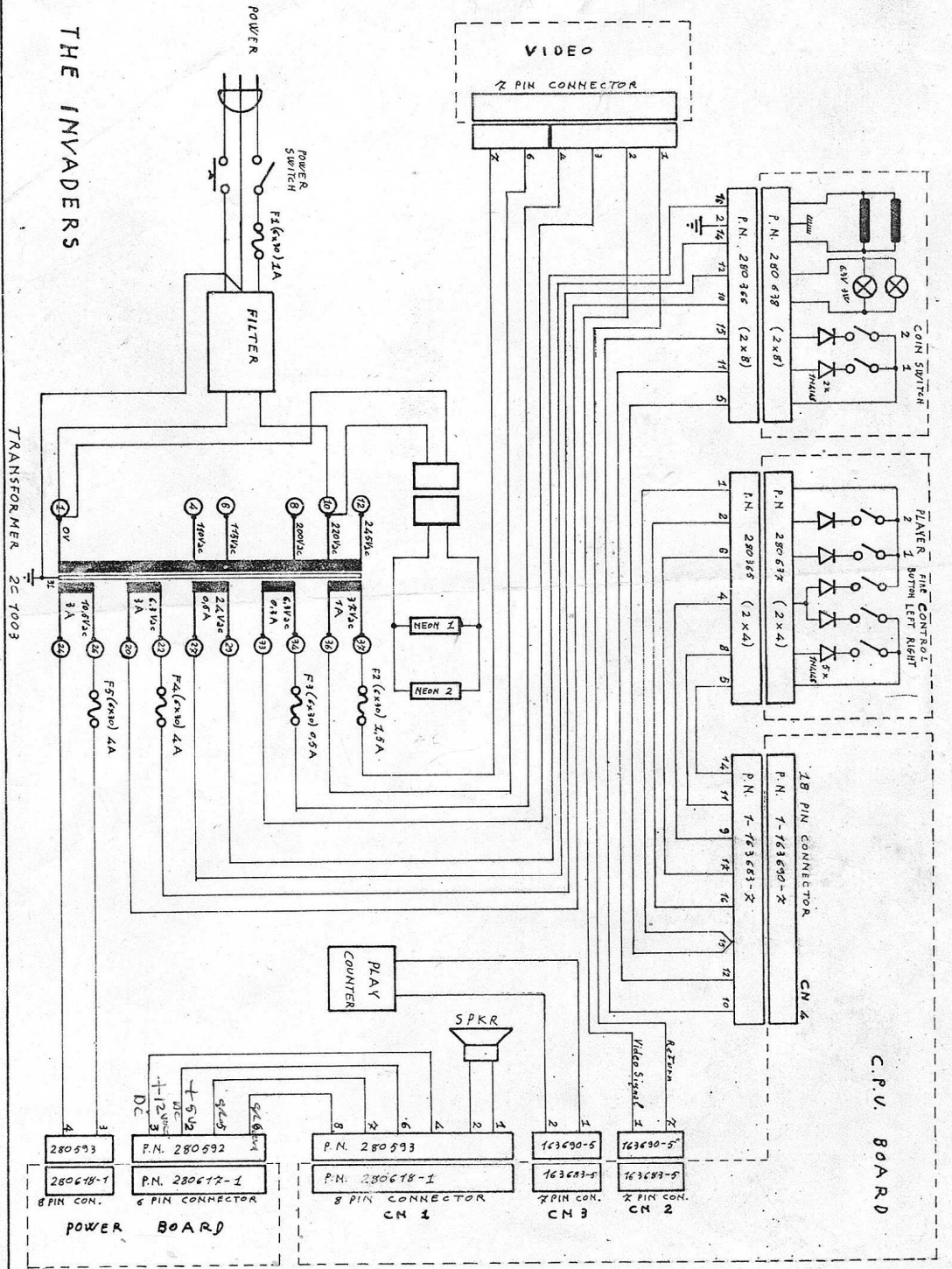
1000 - OFF - 1  
1500 - ON - 1

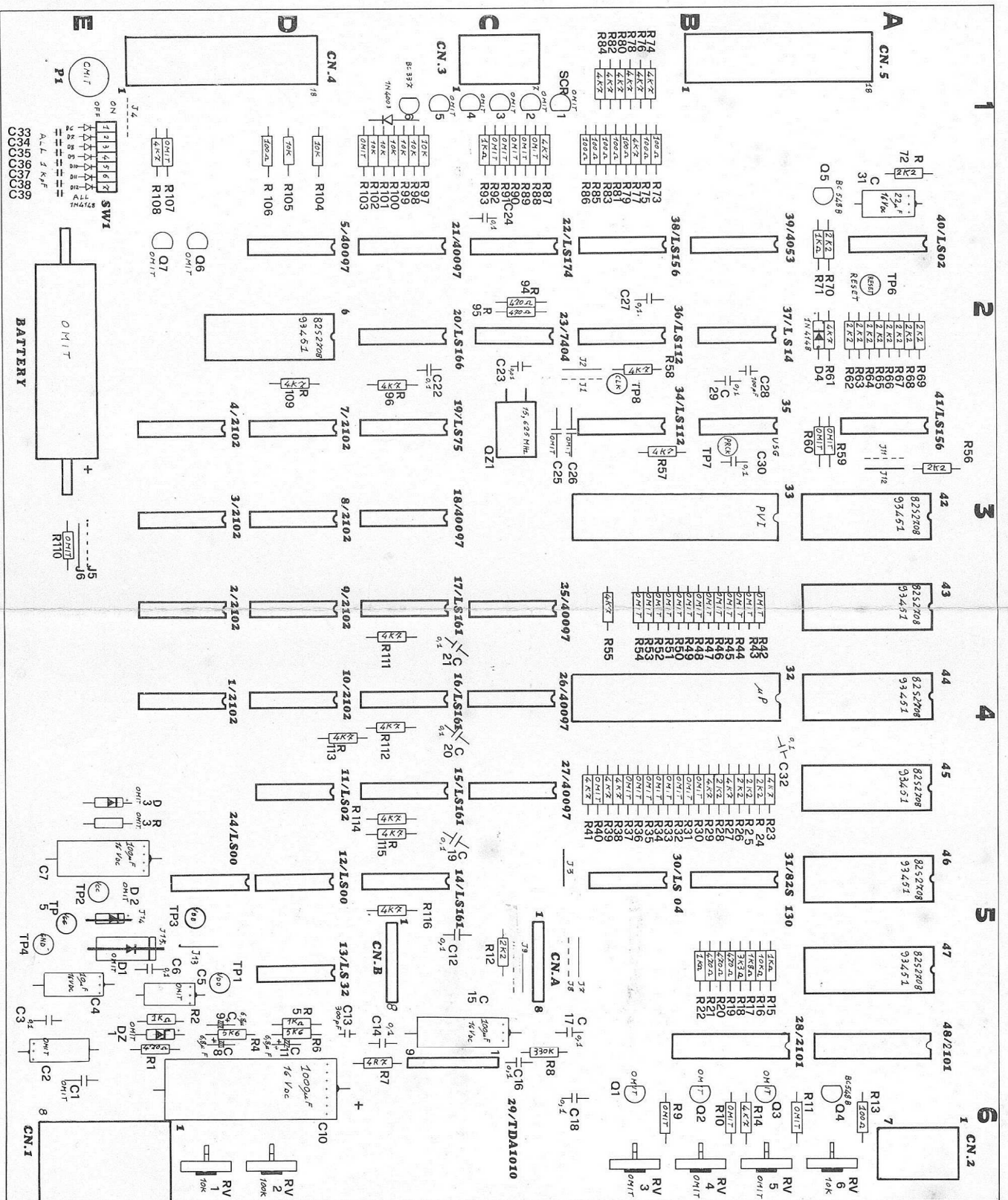
PUNTI SW7 PROLUNGAMENTO GIOCO

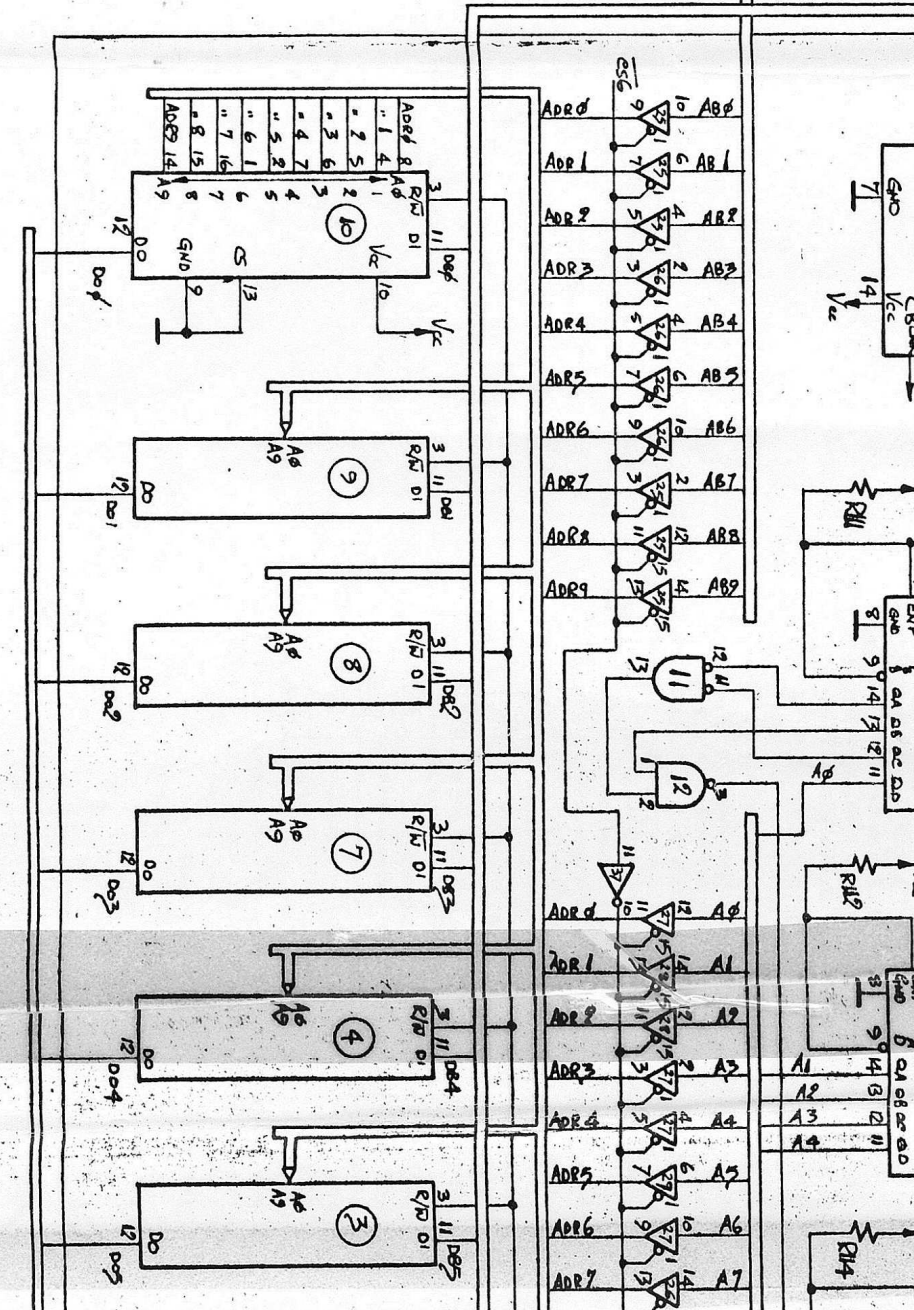
4500 - OFF - NO  
4500 - ON - YES

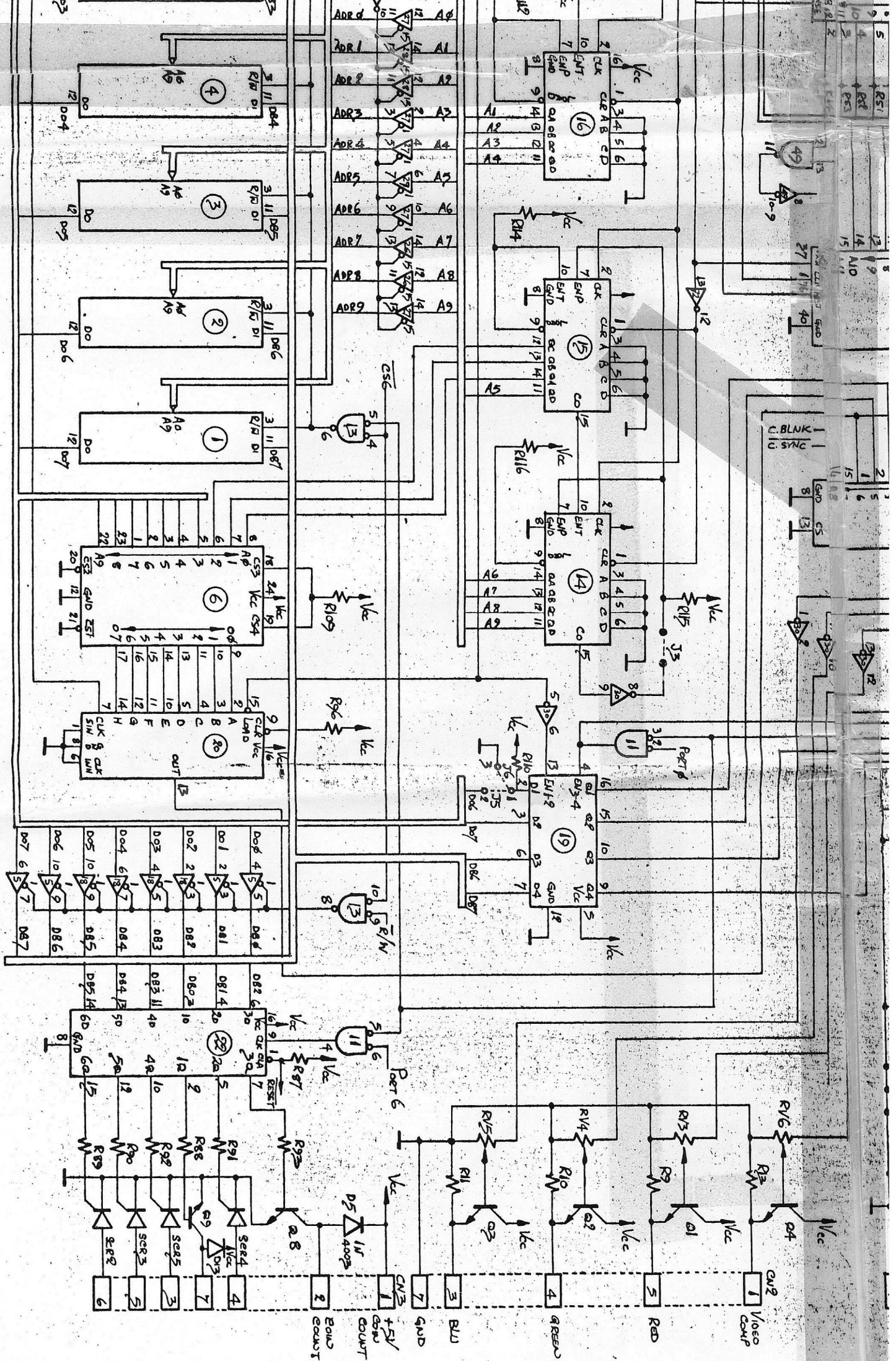
N.B. LA MACCHINA PUO' ESSERE REGOLATA PER UN GIOCO FACILE OPPURE DIFFICILE TRAMITE LA COMUTAZIONE DELLO "SWITCH N° 5". CONSIGLIAMO IL GIOCO FACILE NEL PERIODO INIZIALE (SWITCH 5 IN OFF). IN SEGUITO SI POSSONO AUMENTARE NOTEVOLMENTE LE DIFFICOLTA' DEL GIOCO (SWITCH 5 IN ON) AUMENTANDO LA VELOCITA' DEL PROIETTILE A SPIRALE DEGLI INVASORI.

# THE INVADERS





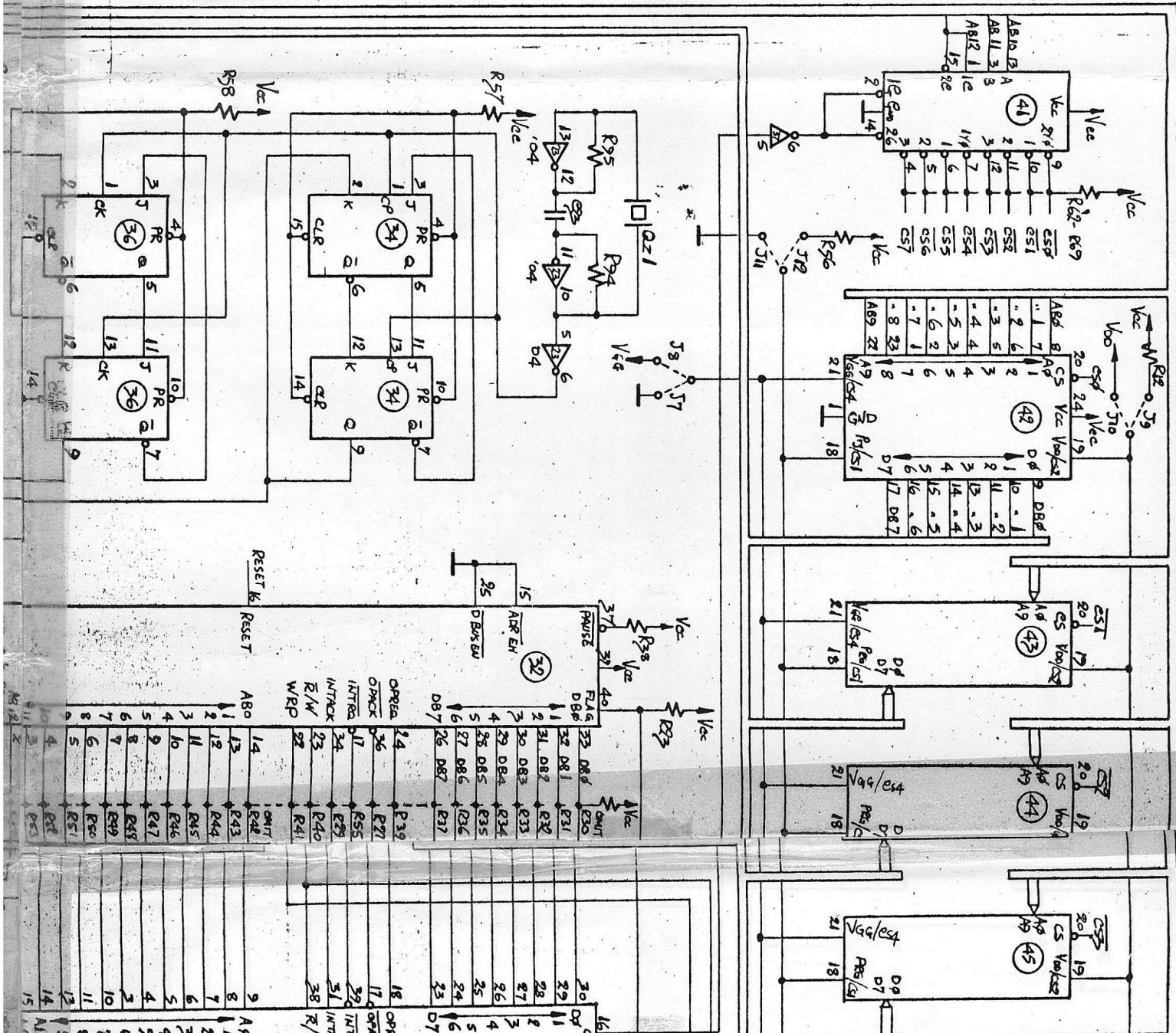


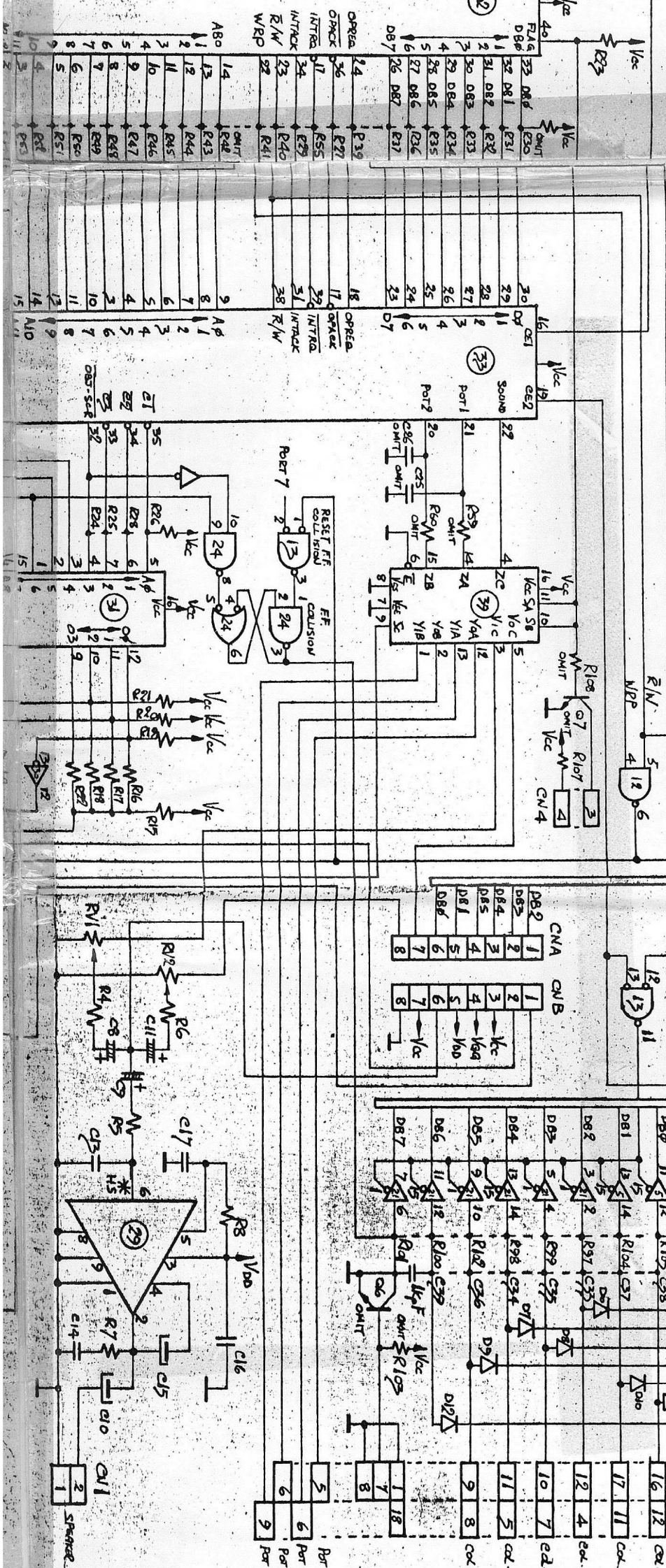


THE INVADERS

ZACCARIA

| ITEM No | SYM                 | NOMENCLATURE OR DESCRIPTION      | QTY |
|---------|---------------------|----------------------------------|-----|
| 1       | PCB                 | PRINTED CIRCUIT BOARD 1B1120     | 1   |
| 2       | IC1-4, IC10         | MOS IC 2102A-4 RAM               | 8   |
| 3       | IC5, 18, 21, 25, 27 | CMOS IC 40037                    | 6   |
| 4       | IC6                 | BIPOLAR IC 93451 PROM            | 1   |
| 5       | IC11, IC40          | LS IC 74LS02 NOR                 | 2   |
| 6       | IC12, IC24          | LS IC 74LS00 NAND                | 2   |
| 7       | IC7                 | LS IC 74LS32 OR                  | 1   |
| 8       | IC4-1, IC7          | LS IC 74LS161                    | 4   |
| 9       | IC19                | LS IC 74LS75                     | 1   |
| 10      | IC20                | LS IC 74LS166                    | 1   |
| 11      | IC22                | LS IC 74LS174                    | 1   |
| 12      | IC23                | TTL IC 7404                      | 1   |
| 13      | IC28, IC48          | MOS IC 2101A-4 RAM               | 2   |
| 14      | IC29                | LINEA IC TDA1010 AUDIO-AMP       | 1   |
| 15      | IC34                | BIPOLAR IC 825130 PRND           | 1   |
| 16      | IC30                | LS IC 74LS04                     | 1   |
| 17      | IC32                | MOS IC 2650A (CT433) $\mu P$     | 1   |
| 18      | IC33                | MOS IC 2436 (CT435) FM           | 1   |
| 19      | IC34, IC36          | LS IC 74LS112 FT                 | 2   |
| 20      | IC35                | MOS IC 2621 (CT430) US6          | 1   |
| 21      | IC32                | LS IC 74LS14                     | 1   |
| 22      | IC38, IC41          | LS IC 74LS156                    | 2   |
| 23      | IC39                | C-MOS IC 4053                    | 1   |
| 24      | Q1-Q7               | TRANSISTOR BC548B                | 7   |
| 25      | Q8                  | TRANSISTOR BC337                 | 1   |
| 26      | SCR-SC85            | SCR 2N5060                       | 5   |
| 27      | D1                  | DIODE 1N5400                     | 1   |
| 28      | D2-D6, D8           | DIODE 1N4148                     | 1   |
| 29      | D5                  | DIODE 1N4003                     | 1   |
| 30      | D21                 | ZEGER DIODE BZX79/CAUT           | 1   |
| 31      | Q21                 | QUARTZ HZ5 15625 MHZ             | 1   |
| 32      | N1, Q1              | N1: Q1 BATTERY CS1003            | 1   |
| 33      | C1, 3, 6, 12, 14    | CERAMIC CAP 0.1 $\mu F$ 50V      | 15  |
| 34      | C2, C4              | ELECTROLYTIC CAP 10 $\mu F$ 16V  | 2   |
| 35      | C3                  | ELECTROLYTIC CAP 22 $\mu F$ 16V  | 1   |
| 36      | C7, C15             | ELECTROLYTIC CAP 100 $\mu F$ 16V | 2   |
| 37      | C8, 9, 11           | TANTALUM CAP 6.8 $\mu F$ 16V     | 3   |
| 38      | C10                 | ELECTROLYTIC CAP 100 $\mu F$ 16V | 1   |
| 39      | C5                  | TANTALUM CAP 10 $\mu F$ 16V      | 1   |
| 40      | C13                 | CERAMIC CAP 300 $\mu F$ 50V      | 1   |
| 41      | C13                 | CERAMIC CAP 10 $\mu F$ 50V       | 1   |
| 42      | C5, C26             | FILM CAP 0.33 $\mu F$ 50V        | 2   |
| 43      | C18, C40            | CERAMIC CAP 100 $\mu F$ 50V      | 2   |
| 44      | C33-C39             | CERAMIC CAP 1K $\mu F$ 50V       | 7   |
| 45      | R1, R2, R4, R5      | CARBON RES 470 $\Omega$ 1/4W 5%  | 6   |
| 46      | R7                  | CARBON RES 1K $\Omega$ 1/4W 5%   | 1   |
| 47      | R3, 9, 10, 11, 13   | CARBON RES 100 $\Omega$ 1/4W 5%  | 14  |
| 48      | R4, 6               | CARBON RES 5K $\Omega$ 1/4W 5%   | 2   |
| 49      | R7                  | CARBON RES 47K $\Omega$ 1/4W 5%  | 1   |
| 50      | R8                  | CARBON RES 220K $\Omega$ 1/4W 5% | 1   |
| 51      | R9                  | CARBON RES 3K3 $\Omega$ 1/4W 5%  | 1   |
| 52      | R9, R10             | CARBON RES 22K $\Omega$ 1/4W 5%  | 2   |





|        |   |             |
|--------|---|-------------|
| R.V.1  | - | CONTRAST    |
| R.V.2. | - | SNAKE LEVEL |
| R.V.3. | - | BRIGHTNESS  |
| R.V.5. | - | BOLP. LIN   |
| R.V.6. | - | HEIGHT      |
| R.V.8. | - | FRAME HOLD  |
| R.V.7. | - | TOP LIN     |

R.V.G. + 30 VOLT REG  
C. 17 - LINE HOLD  
P. 7 - HORIZONTAL HOLD

